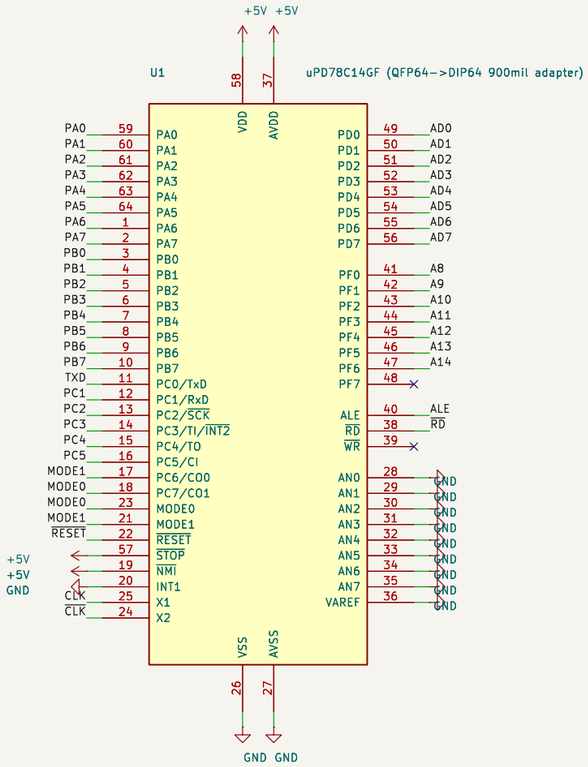
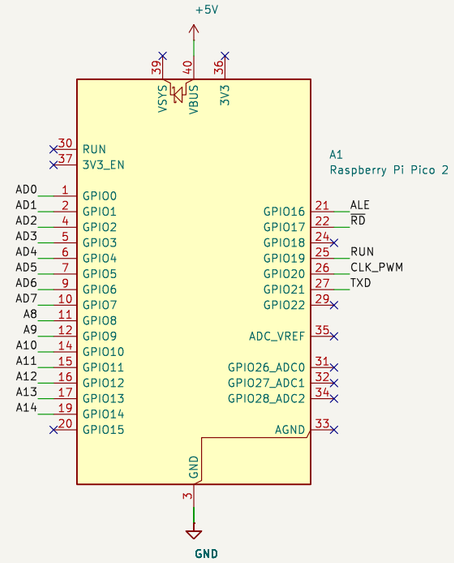
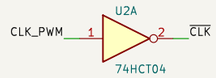
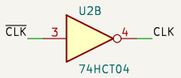
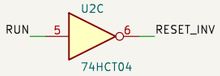
HOW IT WORKS: both MODE pins pulled up -> chip boots from 64 KB external ROM (served by the Pico on AD0–7/A8–14).
The reader code jumps to its 8000h mirror and drives PC7 low; PC7 is tied to MODE0, so the internal
16 KB mask ROM appears at 0000h and is sent as Intel HEX on PC0/TxD at f_clk/96 baud (65104 @ 6.25 MHz).
X1/X2 and /RESET need >= 4 V, hence the 74HCT04 at 5 V. Chip pin numbers are QFP-64 (GF-3BE) pins;
U1's footprint is the 900-mil DIP-64 of a QFP64->DIP64 adapter, pin N = QFP pin N. Pico 2 GPIOs are 5 V tolerant.

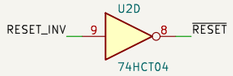


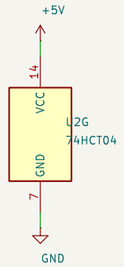


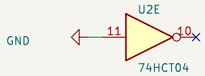


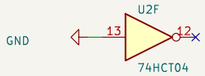


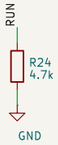


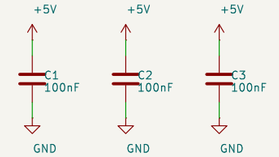




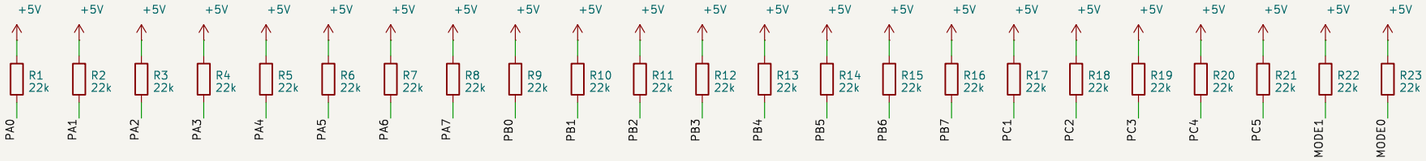








C1 at U1 VDD (58), C2 at U1 AVDD (37), C3 at U2 pin 14



Pull-ups: R1–R8 PA0–7, R9–R16 PB0–7, R17–R21 PC1–5, R22 MODE1/PC6, R23 MODE0/PC7. R24 keeps the chip in reset until the Pico runs.

PC Engine Duo CD controller uPD78C14GF dump
Port of devinacker/upd7811-reader (firmware public domain, hardware CC BY-SA 4.0)

Sheet: /
File: upd78c14-pico-reader.kicad_sch

Title: uPD78C14 mask-ROM reader – Raspberry Pi Pico 2

Size: A3	Date: 2026-10-26	**Rev: 1**
KiCad E.D.A. 9.0.8		Id: 1/1